

Outset · AI in Healthcare · Day 3 · Group 1

Predicting good CRISPR guides

Can a model read 20 letters of DNA and guess how well the molecular scissors will cut?

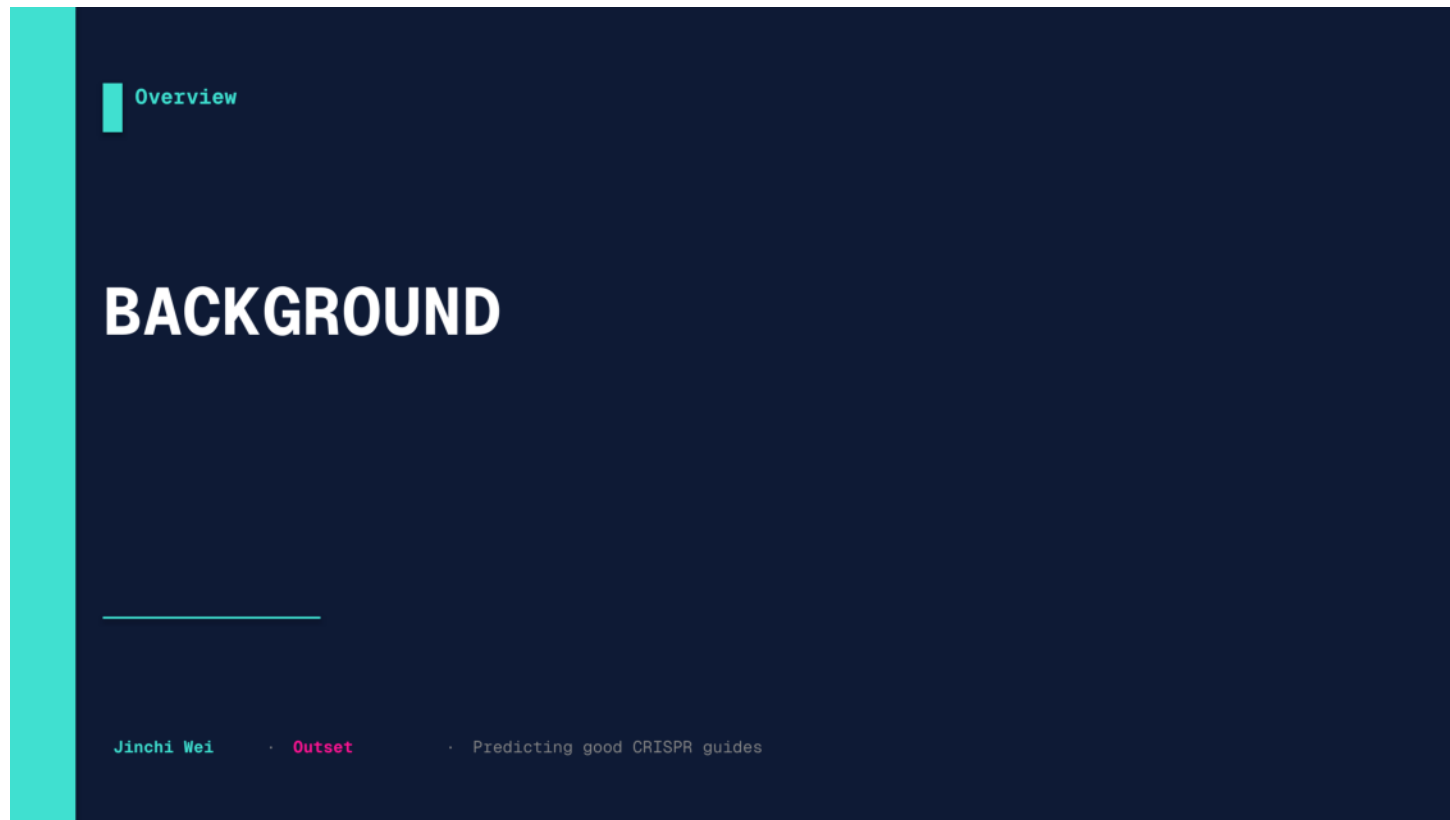
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Outset

2026-07-08



Slide 1 -

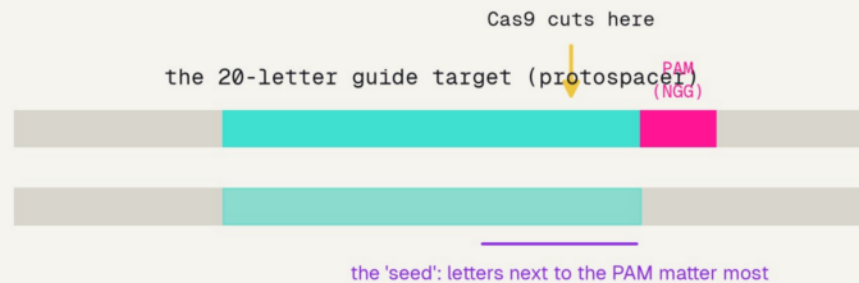


Slide 2 – Background

We open with the biology so the modeling lands later. Three ideas to plant here: CRISPR is a 20-letter guided cut, real scientists SCORE a guide from its sequence instead of testing every one, and the seed region next to the PAM matters most. Transition: let's start with what a guide actually is.

What CRISPR actually is

CRISPR-Cas9 is a pair of molecular scissors you aim at one spot in DNA using a short 20-letter guide (the letters are A, C, G, T). Where the guide matches, the scissors cut -- and that cut turns a gene off or repairs a mutation.



adapted from Hsu, Lander & Zhang 2014

The 20 letters in our dataset ARE this guide sequence. Cas9 locks onto the PAM (the NGG landmark), then cuts a few letters upstream.

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Slide 3 – What CRISPR actually is

Open concrete and physical. CRISPR-Cas9 is scissors plus a 20-letter address label. Point at the figure: the turquoise block is the guide/protospacer, the pink block just to its right is the PAM (always NGG), and Cas9 snips a few letters upstream. Every guide in our dataset is one of these 20-letter strings. Do not mention machine learning yet -- just make sure they can read the picture. Transition: so how do real scientists decide which 20 letters to use?

How real scientists pick a guide

Two guides that look almost identical can behave completely differently -- one cutting beautifully, the other barely at all. Testing each one in the lab takes weeks, so scientists predict cutting efficiency from the 20 letters instead of measuring it.

SCORE, DO NOT EYEBALL

The go-to tool is Doench 2016 Rule Set 2 / Azimuth, which scores a guide straight from its sequence [1], built on the earlier position-and-PAM rules of Rule Set 1 [2].

THE SEED IS KING

Cas9 grabs a landmark called the PAM, then checks the handful of letters right next to it -- the seed. Guides that mismatch there usually fail to cut [3][4].

NEURAL NETS PUSH FURTHER

Newer tools like DeepCRISPR learn the same job with a neural network and far more data, beating hand-built rules [5][6].

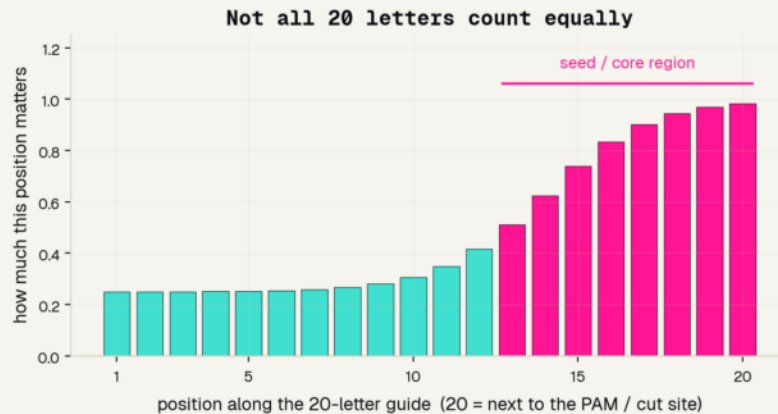
Slide 4 – How real scientists pick a guide

This is the professional context. Real guide selection is a scoring problem, not a guessing game.

Name-drop the three ideas the way the field actually thinks about them: (1) the Doench Rule Set 2 / Azimuth model scores sequence directly [1][2]; (2) the seed region near the PAM dominates [3][4]; and (3) modern neural nets like DeepCRISPR push accuracy further with more data [5][6]. Students should leave knowing the seed idea. Transition: let's look at that seed effect as a picture.

The seed region matters most

Not all 20 letters count the same. The letters nearest the PAM -- the seed, roughly positions 13 to 20 -- carry far more of the signal about whether a guide will cut.



adapted from Zheng et al. 2017 / Doench et al. 2016

READ THE SHAPE

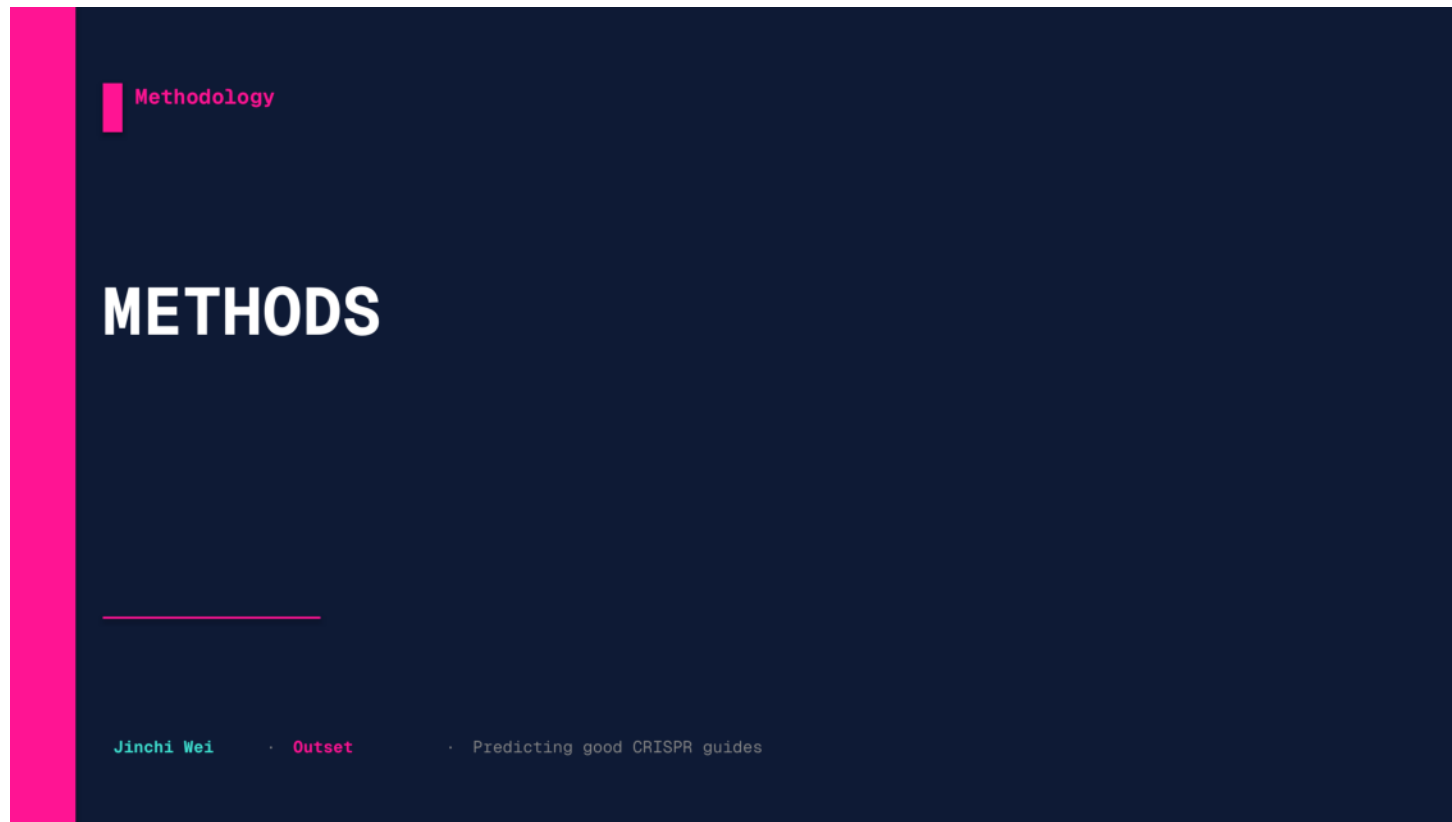
The bars stay low across the far end of the guide, then rise sharply toward the PAM. That rising block is the seed, or core region.

Keep this picture in mind. With no biology lessons at all, our own model is about to rediscover this exact shape.

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Slide 5 – The seed region matters most

Plant the single most important intuition: the 20 letters are not equal. The chart rises toward the PAM end -- that block is the seed. This is a redraw of the Zheng 2017 / Doench 2016 finding [4][1]. Tell them to remember the shape, because the payoff slide later shows OUR model reproducing it from scratch. Transition: now to our actual project -- the task and the data.



Slide 6 – Methods

Now we turn that biology into a machine-learning task. Coming up: the exact dataset, the class imbalance, and the single decision that dominates everything -- how we turn DNA letters into numbers. Keep the seed idea handy, because it drives the prediction we are about to make.

The question we are asking

We use the exact dataset behind the famous Rule Set 2 model [1]: real guides that scientists actually measured in the lab. Our job is to predict, from the 20 letters alone, whether a guide is a strong cutter.

5,310

real 20-letter guides, each with a measured cutting score from 0 to 1

17

different human genes, so the model must learn sequence rules, not memorize one gene

~20%

of guides cut well, so a lazy "always low" guesser looks 80 percent accurate and is useless

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Slide 7 – The question we are asking

Set up the task in numbers. 5,310 real, lab-measured guides across 17 genes -- the same data behind the Rule Set 2 model [1]. The number to dwell on is the last one: only about 1 in 5 guides is a good cutter. That class imbalance is why, two slides from now, we refuse to trust plain accuracy. Say it out loud: an 'always low' model would be 80 percent accurate and completely useless. Transition: the first real decision -- how do we hand DNA letters to a model?

Turning letters into numbers

A computer cannot do math on the letter A, so we turn each guide into numbers -- and how we do that is the single most important choice in the whole project. We tried two ways and let the results decide.

ONE-HOT (keeps the order)

Four on/off switches per position, so the model always knows which letter sits where. That is 20 positions times 4 = 80 numbers per guide.

K-MER (ignores the order)

Just counts how many of each letter and letter-pair appear, throwing the position away. Only 21 numbers per guide.

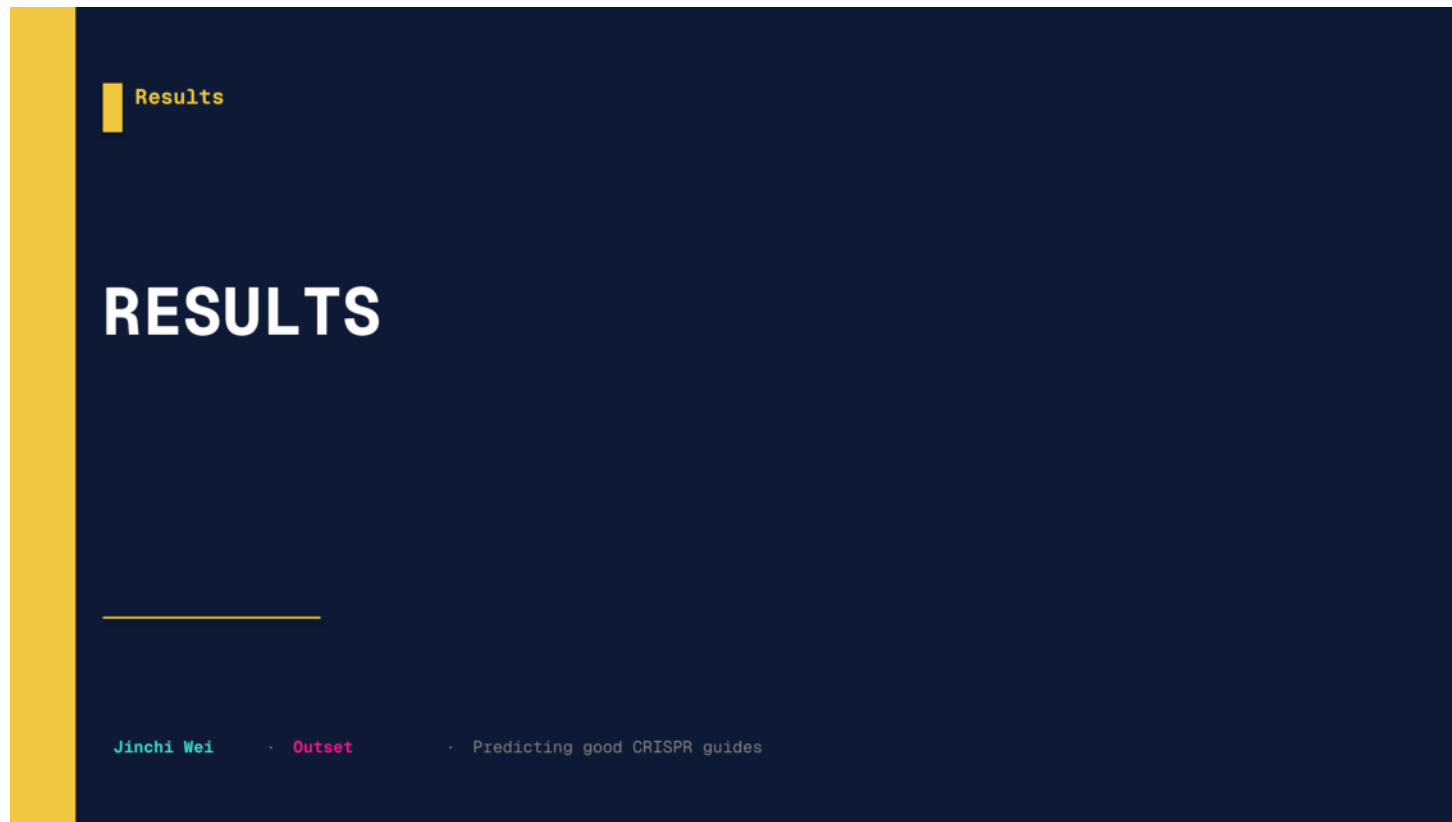
LISTEN vs SILENT

Same letters, different words. k-mer cannot tell them apart, but one-hot can -- and since biology cares exactly WHERE a letter sits (the seed!), one-hot should win.

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Slide 8 – Turning letters into numbers

The conceptual heart of the project. Featurizing = turning letters into numbers, and the choice encodes what you think matters. one-hot keeps position (80 numbers); k-mer keeps only counts (21 numbers) and throws order away. The LISTEN/SILENT analogy is the whole lesson in two words: same letters, different meaning -- order matters. Because the seed is a POSITION effect, we predict one-hot wins. Make them commit to that prediction before the next slide. Transition: let's test it.

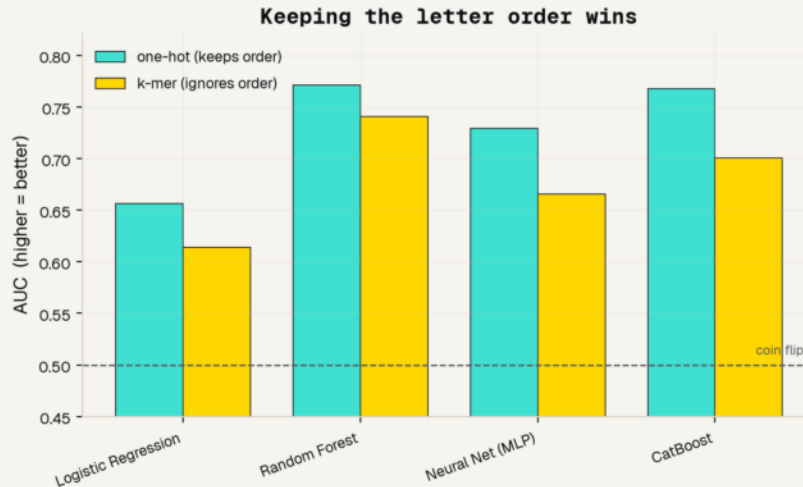


Slide 9 – Results

Time for what we found. Three beats here: keeping the letter order (one-hot) wins, why we grade with AUC instead of accuracy, and the payoff -- the model rediscovering the seed on its own. Then a harder task, predicting the exact cutting score.

Keeping the letter order wins

We trained the same four models on each way of making numbers and compared them. Every single model did better with one-hot than with k-mer.



THE REAL LESSON

The one-hot bars (turquoise) top the k-mer bars (gold) for every one of the four models. How you prepared the data mattered more than which fancy model you picked.

0.77

best AUC -- Random Forest on one-hot

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Slide 10 – Keeping the letter order wins

First result, and the transferable one. Same four models, two featurizations; one-hot wins across the board. The takeaway is bigger than CRISPR: data representation beat model choice. The prediction they made last slide came true, and for the right reason -- one-hot kept the position information where the signal lives. Best score is Random Forest on one-hot at AUC 0.77. Transition: wait, why AUC and not just accuracy?

Why we grade with AUC, not accuracy

Because only about 20 percent of guides cut well, plain accuracy is easy to fake. AUC asks a fairer question.

80%

accuracy of a lazy model that ALWAYS guesses "low" -- and never finds a single good guide. Accuracy is fooled by the imbalance.

0.77

AUC of our best model. AUC asks a fairer question: does it rank GOOD guides above bad ones? 0.5 is a coin flip, 1.0 is perfect.

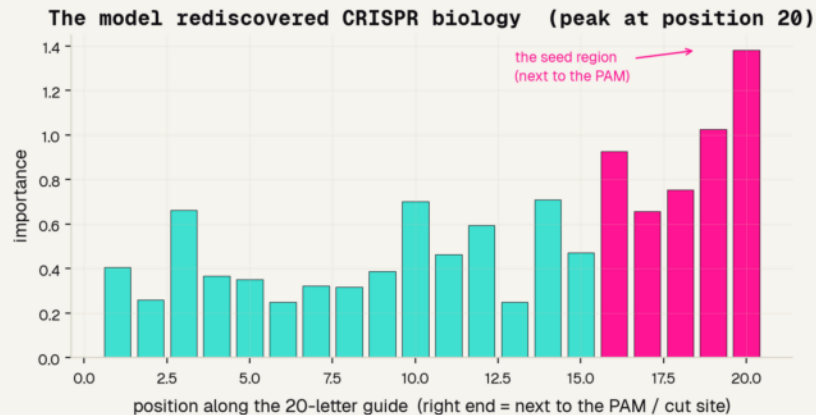
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Slide 11 – Why we grade with AUC, not accuracy

The judgment slide. On imbalanced data accuracy lies: an 'always low' model scores 80 percent and is worthless because it never flags a usable guide. AUC sidesteps the trick -- it measures whether the model RANKS good guides above bad ones, so 0.5 is a coin flip and 1.0 is perfect. Our 0.77 means real ranking skill. This same trap (accuracy on imbalanced data) shows up everywhere in medical AI, so it is worth over-teaching. Transition: now the most satisfying result.

The model rediscovered the seed

Now the payoff. We asked a simple, explainable model which positions along the guide it leaned on. Nobody ever told it about seeds.



WHAT IT FOUND

Its importance climbs toward the right end of the guide and peaks at position 20, right next to the PAM. That is the seed region -- exactly where the biology said the action is.

The model learned real CRISPR biology from nothing but sequences and cut-scores. A model that agrees with known science is one people will trust.

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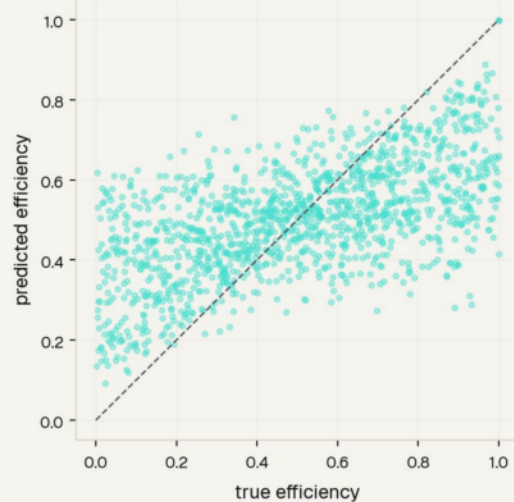
Slide 12 – The model rediscovered the seed

The wow moment -- put it up and pause. We asked a plain logistic-regression model which of the 20 positions it relied on, and the importance peaks at position 20, right at the PAM: the seed. Flip back to the background seed chart -- same shape. Nobody encoded that; the model recovered decades of CRISPR biology from sequences and cut-scores alone. Tie it to trust: an explainable model that matches known science is one clinicians and biologists will believe. Transition: can it predict the exact score, not just high vs low?

Predicting the exact score

High versus low throws away detail, so we also tried to predict the exact 0-to-1 cutting score. R-squared says how much of the real variation we explained.

Predicting the exact score (R^2 0.36, r 0.61)



0.36

R-squared: the share of the ups-and-downs the model explained (0 = guessing the average, 1 = perfect).

0.61

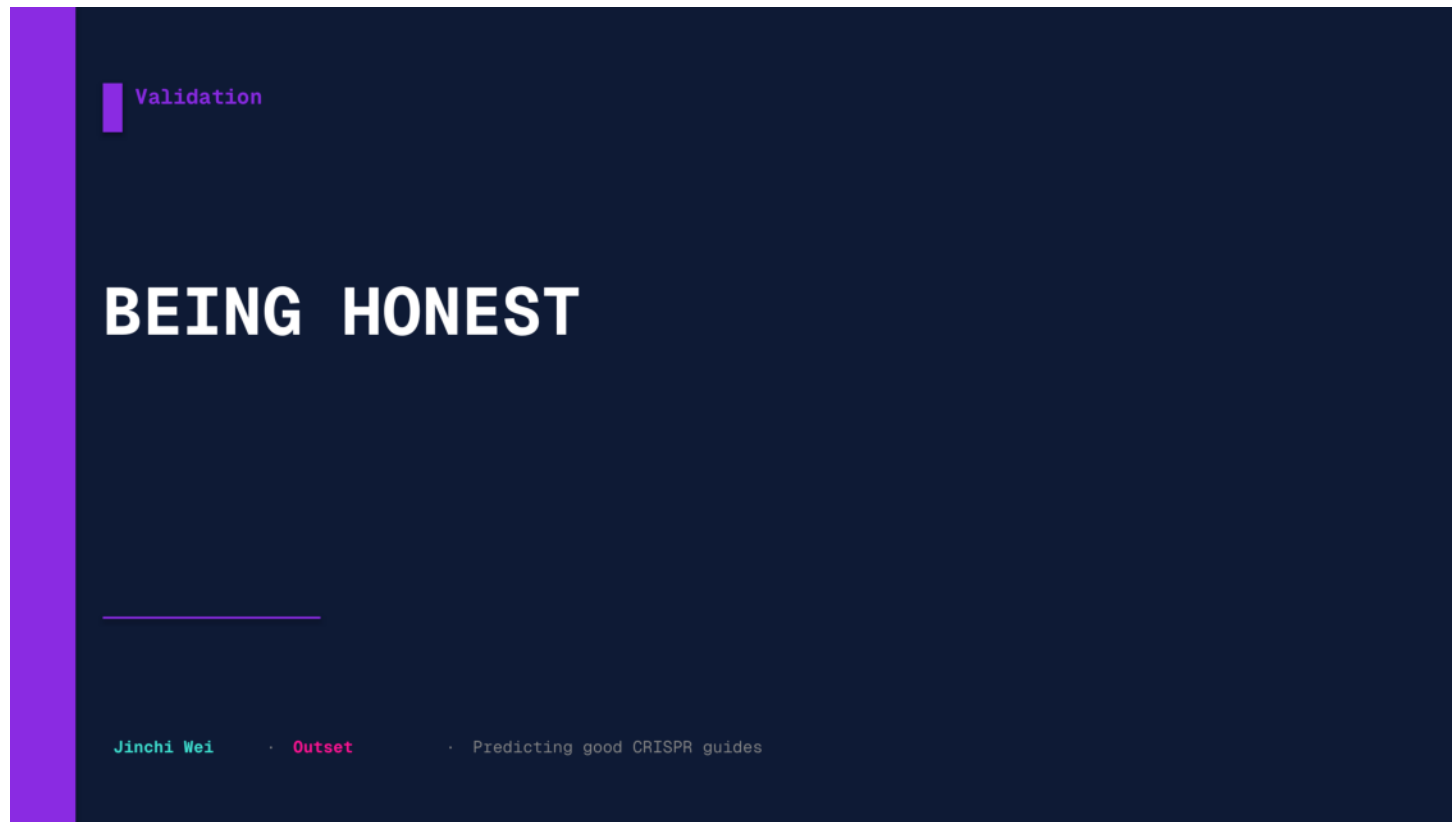
correlation between predicted and true efficiency.

Real signal, but far from perfect -- and that is honest. The pro tools scatter the same way [1][5][6].

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Slide 13 – Predicting the exact score

Second, harder task: predict the exact 0-to-1 score, not just a yes/no. Explain R-squared plainly -- the fraction of the real variation the model captures, 0 is guessing the mean, 1 is perfect. We get 0.36 (correlation 0.61): a real but loose trend, visible as a tilted cloud around the diagonal. The honest point: the professional predicted-vs-measured plots scatter just like ours [1][5][6]. Useful does not mean perfect. Transition: so let's be clear-eyed about what this toy can and cannot do.



Slide 14 – Being honest

Good science states its own limits out loud. We lay out what this toy model genuinely shows, what it cannot do, and the off-target-safety question it says nothing about. Then the references and the one-sentence takeaway. This is the section the rubric rewards most.

What this toy model can and cannot do

A good project names its own limits out loud. Ours is a demonstration of an idea, not a design tool -- and here is exactly where the line sits.

IT CAN

Show that a guide's sequence really carries signal, and that the seed region matters -- on the very same data the professionals use [1].

IT CANNOT MATCH THE PROS

It will not beat a production tool, and it ignores the cell-type and chromatin effects that change cutting inside a living cell [7].

IT CANNOT JUDGE SAFETY

It says nothing about off-target risk -- whether the guide also cuts the wrong places elsewhere in the genome [8].

Slide 15 – What this toy model can and cannot do

Model the intellectual honesty the rubric rewards. Green light: it genuinely shows sequence carries signal and the seed matters, on the pros' own data [1]. Red lights: it will not beat a production tool and cannot see cell-type or chromatin effects [7]; and crucially it says NOTHING about off-target safety [8] -- the single most important question for real CRISPR use. State plainly: this builds intuition, it does not choose a clinical guide. Transition: the papers, then the bottom line.

References

The eight papers behind this project -- from the plain-English CRISPR reviews to the models we measured ourselves against.

FOUNDATIONS

- [1] Doench et al. 2016, Nat Biotechnol -- Rule Set 2 / Azimuth; source of our dataset.
- [2] Doench et al. 2014, Nat Biotechnol -- Rule Set 1; position + PAM rules.
- [3] Hsu, Lander & Zhang 2014, Cell -- landmark plain-English CRISPR review.
- [4] Zheng et al. 2017, Sci Rep -- direct evidence the seed / core matters most.

MODERN MODELS & REVIEWS

- [5] Chuai et al. 2018, Genome Biol -- DeepCRISPR, a CNN guide-efficiency model.
- [6] Kim et al. 2019, Sci Adv -- DeepSpCas9, trained on 12,832 sites.
- [7] Konstantakos et al. 2022, NAR -- survey comparing guide predictors.
- [8] Abbaszadeh & Shahlai 2025, arXiv -- explainable AI + off-target safety.

Slide 16 – References

Give credit and show the trail. Left column is the foundation: the two Doench Rule Sets [1][2], the Hsu/Lander/Zhang review [3], and the Zheng seed paper [4]. Right column is the modern neural-net line and the surveys [5][6][7][8]. Do not read every line -- point out that [1] gave us the data, [4] is the seed evidence our model reproduced, and [8] is the off-target-safety gap we cannot address. Transition: the one sentence to remember.

The honest bottom line

A model you can explain is a model people trust.

Use this project to build intuition -- then use a validated tool and the lab to actually choose a guide.

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Slide 17 – The honest bottom line

Land the whole deck on one idea: explainability builds trust. Our toy model earned a little trust by agreeing with known biology and by being honest about R-squared 0.36 and about off-target safety. Leave them with the working rule for all of medical AI: a simple model you can explain and honestly evaluate beats a fancy black box nobody can check. Then use the real tools and the lab to actually pick a guide. Thank them.

Thank You

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Slide 18 -